Customer Churn Prediction Using Machine Learning

1. Abstract

Customer churn prediction helps telecom companies identify customers who are likely to leave their services. This project uses Machine Learning to predict customer churn based on customer information such as tenure, contract type, monthly charges, and payment method. A Random Forest Classifier was used to build the prediction model and a Streamlit web application was developed for user interaction.

2. Introduction

Customer churn is one of the major challenges faced by telecom companies. Predicting customer churn helps companies take preventive measures and improve customer retention. This project aims to develop a machine learning model that predicts whether a customer will stay or churn.

3. Problem Statement

To develop a machine learning model that predicts whether a telecom customer is likely to churn based on historical customer data.

4. Dataset Description

Dataset: Telco Customer Churn Dataset. Features include gender, Senior Citizen, Partner, Dependents, Tenure, Phone Service, Multiple Lines, Internet Service, Online Security, Online Backup, Device Protection, Tech Support, Streaming TV, Streaming Movies, Contract, Paperless Billing, Payment Method, Monthly Charges, and Total Charges. Target Variable: Churn (Yes/No).

5. Data Preprocessing

Loaded the dataset, converted TotalCharges to numeric values, removed missing values, dropped customerID, encoded Churn (Yes=1, No=0), and applied Label Encoding to categorical columns.

6. Methodology

Training Data: 80%, Testing Data: 20%. Algorithm Used: Random Forest Classifier.

7. Model Training

The Random Forest Classifier was trained using the processed dataset after train-test split.

8. Model Evaluation

Insert your model accuracy and screenshot here.

9. Streamlit Web Application

A Streamlit application was developed to allow users to enter customer details and predict customer churn.

10. Results

The model successfully predicts whether a customer is likely to stay or churn based on customer information.

11. GitHub Repository

Add your GitHub repository URL here.

12. Live Application

https://customer-churn-prediction-as59ambkdntn8jjfsnkodj.streamlit.app/

13. Conclusion

This project successfully developed a machine learning-based customer churn prediction system using the Random Forest algorithm and deployed it using Streamlit.

14. References

Python Documentation, Scikit-learn Documentation, Streamlit Documentation, Kaggle Telco Customer Churn Dataset.